

Farrell Kimble

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PROJECT WORK EXPERIENCE

Orbit: Parking Traffic and Automation App | 1st Place in Idea and Innovation at UMBC Sept 2025 – Present

- Engineered a normalized **SQL** data model for parking transactions supporting real-time availability tracking.
- Implemented backend validation and reporting logic to detect anomalies, ensure data integrity, and **track KPI metrics**.
- Processed commuter traffic datasets using **Python (Jupyter Notebook)**, identifying peak congestion windows (9am–2pm) to inform system optimization and decision logic.

Full-Stack | Desk Master Schedule | UMBC Department of IT Sept 2025 – Present

- Developed a full-stack scheduling application, implementing frontend interfaces with **JavaScript, HTML, and CSS** and backend logic to manage shift assignments, coverage requests, and real-time updates all within an **AWS Database**.
- Designed and integrated a **relational database (SQL)** to store scheduling data, enabling efficient querying, data validation, and consistent state management across users.
- Implemented application logic to handle shift swaps, conflict detection, and data synchronization, improving system reliability and ensuring accurate schedule updates.

Predictive Analytics Data | Emissions Decision Support System | UMBC Department of IT Sept 2025 – Dec 2025

- Developed Python-based ETL pipelines using **pandas** to extract, transform, and standardize greenhouse gas emissions data across multiple Maryland counties.
- Cleaned and processed large datasets using Python and Excel to align inconsistent data ranges, ensuring accurate comparison.
- Developed interactive dashboards and visualizations in Power BI to communicate insights to non-technical stakeholders.

Data Management Project Lead | Audio Storage System | UMBC Department of IT Sept 2024 – Dec 2024

- Engineered ETL pipelines to extract and transform 100+ records from **Microsoft Access** and load into an **Oracle relational database**, ensuring 100% data integrity.
- Implemented a relational database in Microsoft Access and migrated schemas to Oracle, ensuring data integrity across systems.
- Applied relational database normalization to reduce redundancy, optimize schema design, and improve query performance.

WORK EXPERIENCE

Teaching Fellow IS 296: Foundations of Data Science | UMBC Department of IT January 2026 – Present

- Developed and demonstrated **Python**-based data processing workflows using **pandas in Jupyter Notebook**, guiding 40+ students through implementing data cleaning logic, exploratory analysis functions, and reproducible scripts on real datasets.
- Reviewed and debugged 40+ student **Jupyter Notebook** projects, improving code reliability, data handling accuracy, and documentation through structured feedback on **Python** implementations.
- Analyzed student-generated outputs using **Python and Excel**, identifying recurring logic errors and inefficiencies in data processing pipelines to improve overall system consistency and performance.

Research Intern | AI Sense | UMBC Department of IT Feb 2025 – Aug 2025

- Engineered an embedded system using **Raspberry Pi and Arduino**, implementing computer vision detection logic and automated sound-trigger responses for real-time wildlife deterrence.
- Executed system testing across multiple iterations with 10–12 collaborators, debugging hardware-software interactions and refining performance based on collected system data.
- Built data visualization pipelines in **Power BI**, processing and analyzing system performance metrics to evaluate detection accuracy and system effectiveness.

EDUCATION

University of Maryland, Baltimore County Estimated Grad May 2026

- Major: Information Systems, Minor: Psychology, Business Analyst Certification

Prince George's Community College May 2020 - May 2022

- Computer Programming and Database Systems Certification

Honors: Data Science & AI Scholar, France-Merrick Scholar

Certifications: AI4ALL: Discover AI Certification, Microsoft Access Certification, Google Data Analytics Certification

Coursework: Advanced Oracle and PL(SQL), Computer Science I & II (Python & C++ respectively)

SKILLS

Developer Tools: Visual Studio Code, PyCharm, CLion, Git

Technologies: Linux/Unix, Oracle, Power BI, Tableau, Jupyter Notebook,

Programming Languages: Python, C++, SQL, JavaScript, HTML, CSS,

Concepts: Object-Oriented Programming (OOP), Data Structures, ETL Pipeline Development, Relational Database Design (3NF)